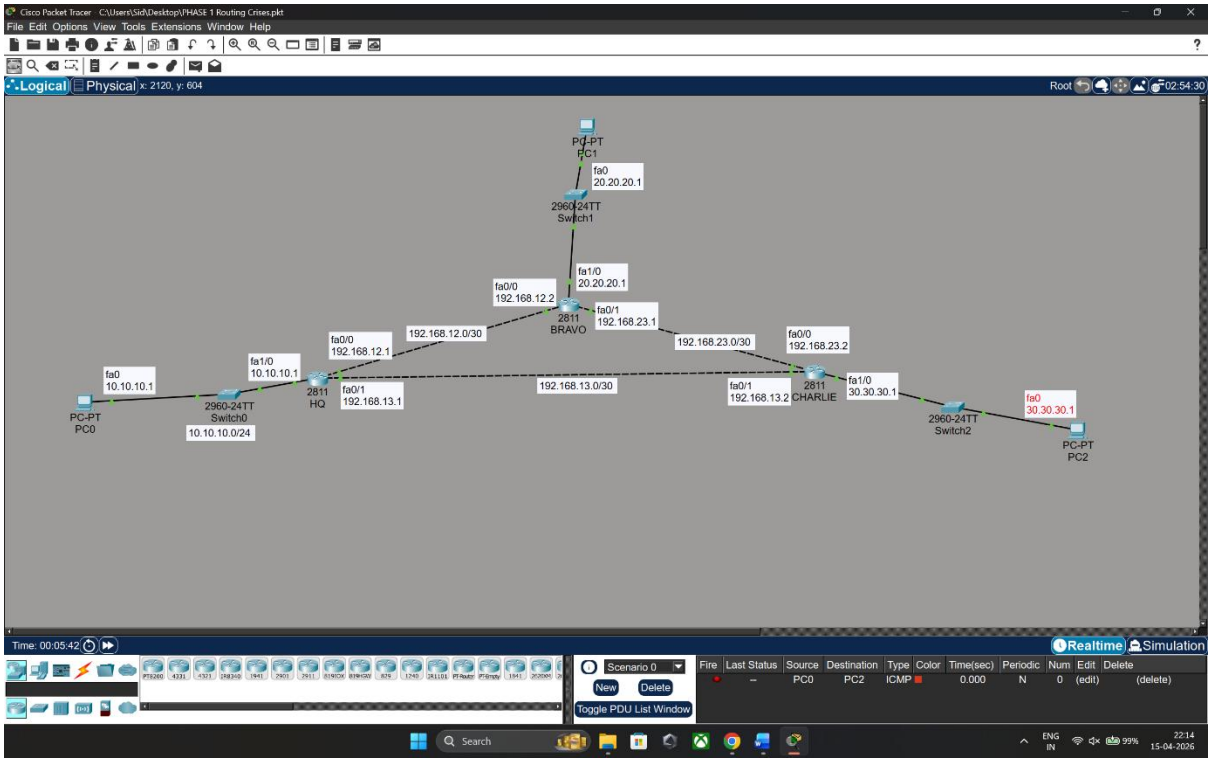
Phase 1 – Routing Crisis (Cisco Packet Tracer)

1. TOPOLOGY



2. CONFIG ROUTERS

- **Router 1(HQ)**

```
ip ospf 1 area 0

----------------
en
config t
hostname R1

int f0/0
 ip add 192.168.12.1 255.255.255.252
 no sh
int f0/1
 ip add 192.168.13.1 255.255.255.252
 no sh
int f1/0
 ip add 10.10.10.1 255.255.255.0
 no sh


router ospf 1
 network 192.168.12.0 0.0.0.3 area 0
 network 10.10.10.0 0.0.0.255 area 0

 redistribute eigrp 100 subnets

router eigrp 100
 network 192.168.13.0 0.0.0.3
 network 10.10.10.0 0.0.0.255
 no auto-summary

 redistribute ospf 1 metric 10000 100 255 1 1500
```

- **Router 2(BRAVO)**

```
en
config t
hostname R2


int f0/0
 ip add 192.168.12.2 255.255.255.252
 no sh
int f0/1
 ip add 192.168.23.1 255.255.255.252
 no sh
int f1/0
 ip add 20.20.20.1 255.255.255.0
 no sh


router ospf 1
 network 192.168.12.0 0.0.0.3 area 0
 network 20.20.20.0 0.0.0.255 area 0

 redistribute eigrp 100 subnets

router eigrp 100
 network 192.168.23.0 0.0.0.3
 network 20.20.20.0 0.0.0.255
 no auto-summary

 redistribute ospf 1 metric 10000 100 255 1 1500
```

- **Router 3 (CHARLIE)**

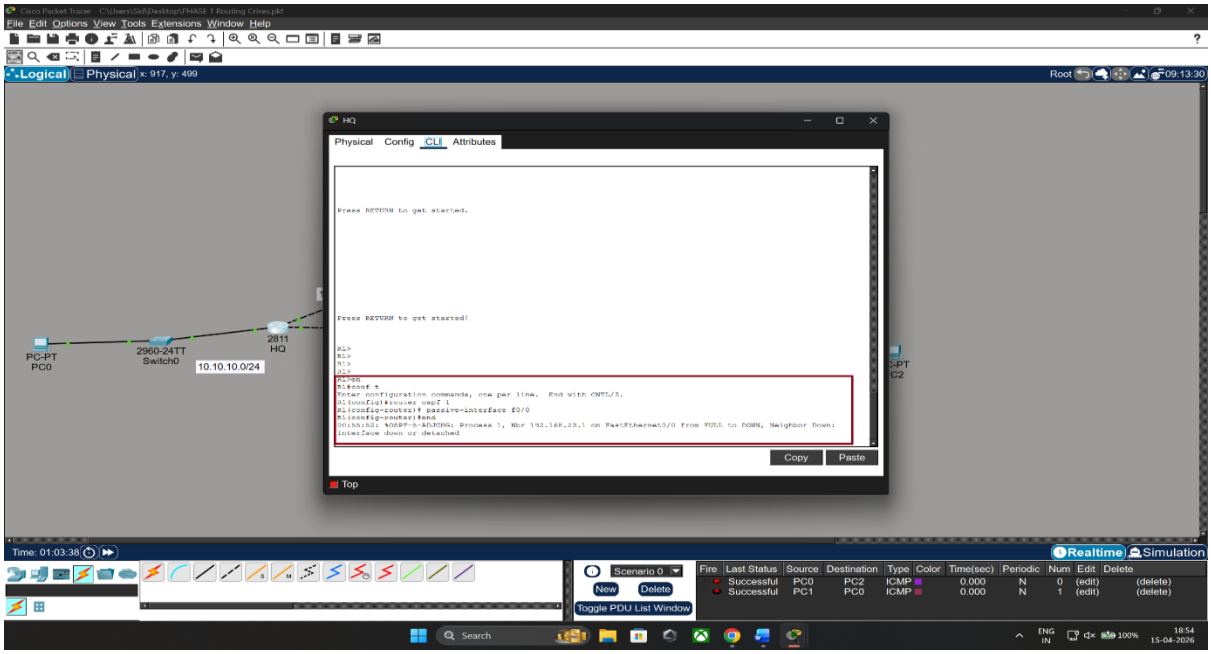
```
!R3 Configuration

en
config t
hostname R3

! -- Interfaces --
int f0/0
 ip add 192.168.23.2 255.255.255.252
 no sh
int f0/1
 ip add 192.168.13.2 255.255.255.252
 no sh
int f1/0
 ip add 30.30.30.1 255.255.255.0
 no sh

router eigrp 100
 network 192.168.23.0 0.0.0.3
 network 192.168.13.0 0.0.0.3
 network 30.30.30.0 0.0.0.255
 no auto-summary
```

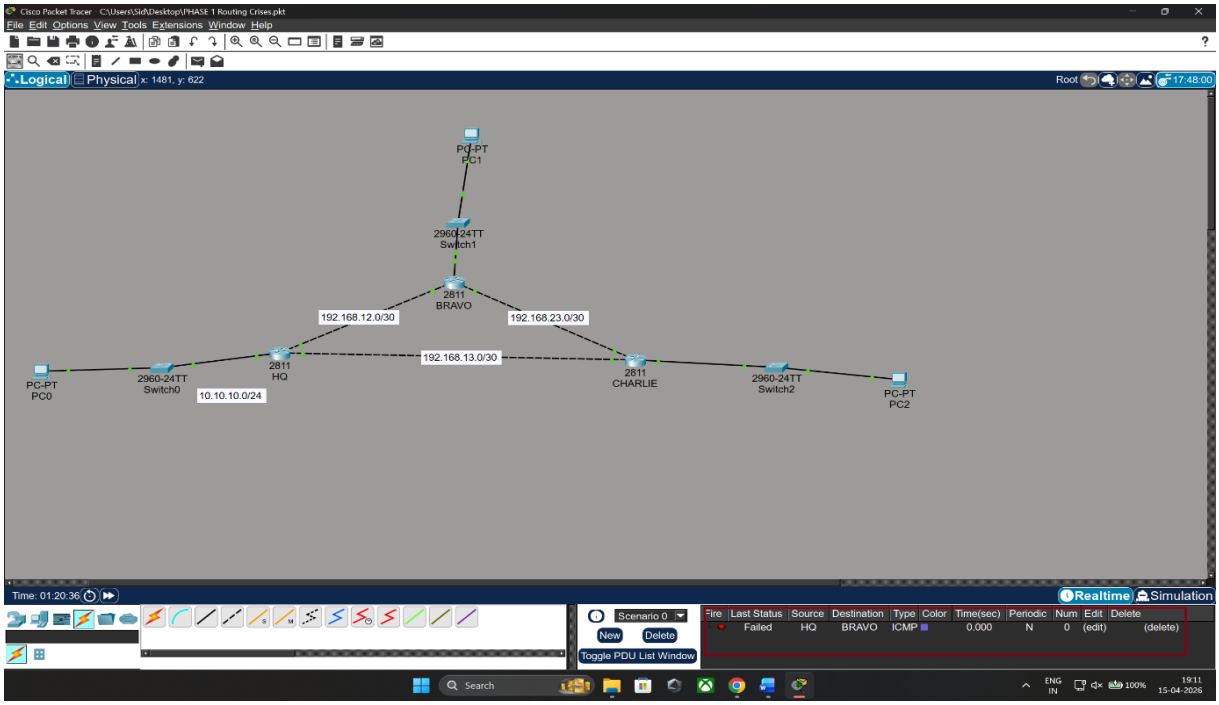
3. REDISTRBUTE –

- The redistribute command is used to share routing information between different routing protocols (e.g., OSPF to EIGRP), allowing routers to learn paths across diverse network domains. Without it, a router running one protocol remains "blind" to the networks managed by another, causing communication gaps between different parts of the organization. It acts as a protocol translator, converting the unique metrics (like hop count or bandwidth) of one protocol into a format the other can understand and process. This is essential for network integration during mergers, migrations, or when connecting specialized tactical equipment to a broader strategic backbone.

4. DELIBERATE FAULT- PASSIVE INTERFACE

A passive interface is a configuration used in routing protocols (like OSPF or EIGRP) that prevents the router from sending or receiving routing updates on a specific interface while still including that network in its routing table. It is a critical security measure used to reduce unnecessary traffic and prevent unauthorized routers from learning or manipulating network topology.

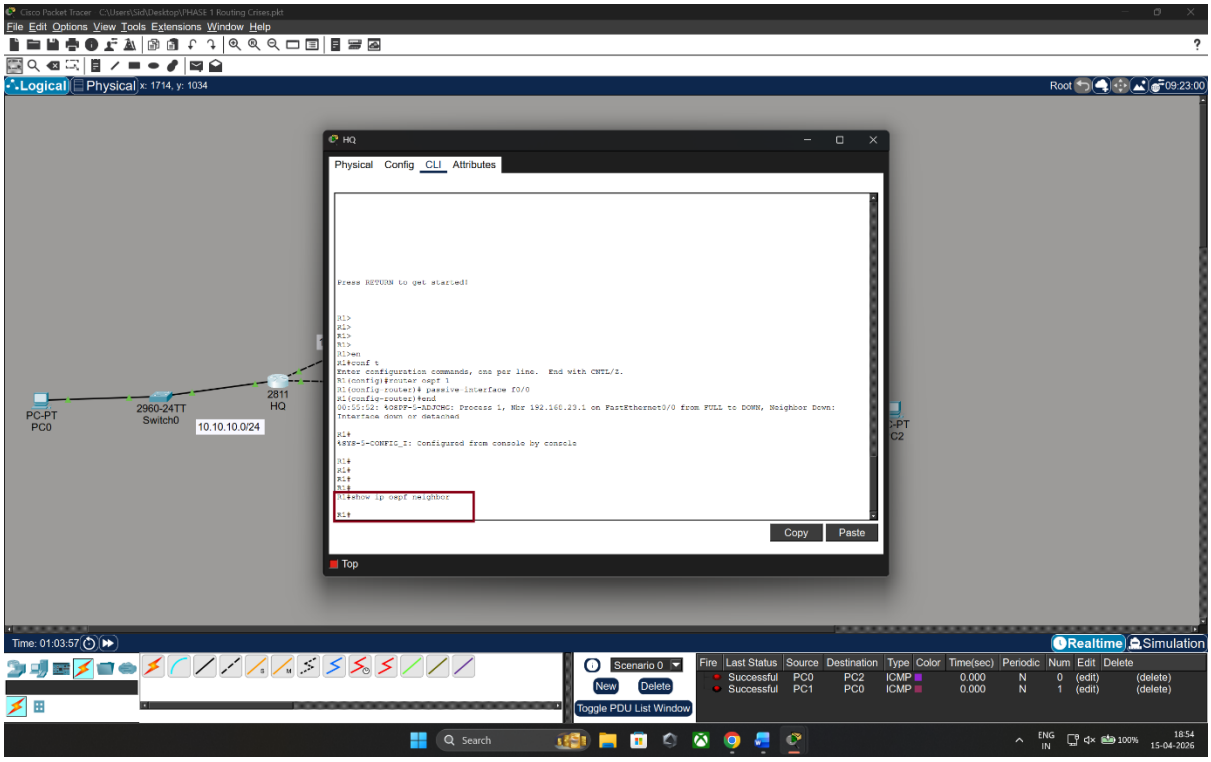




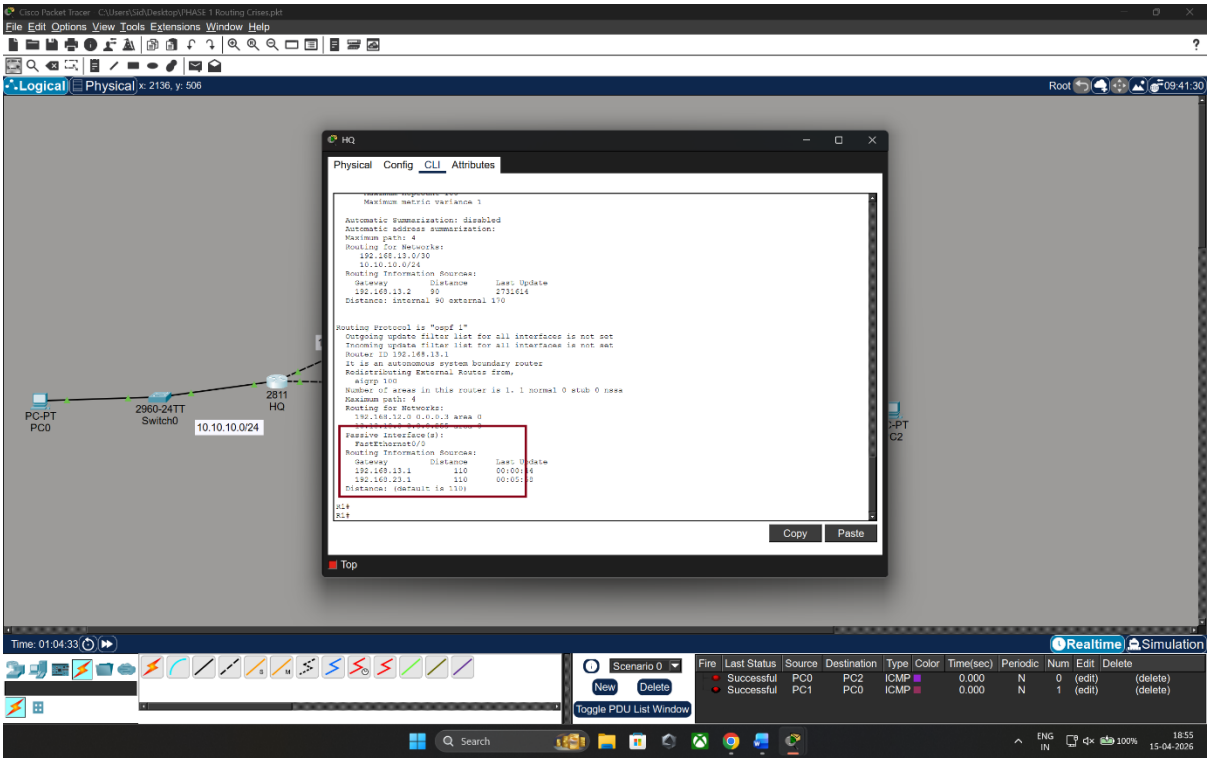
5. Troubleshoot and restore connectivity.

- To identify why the HQ sector could no longer communicate with Bravo, I performed the following diagnostic commands on Router R1-

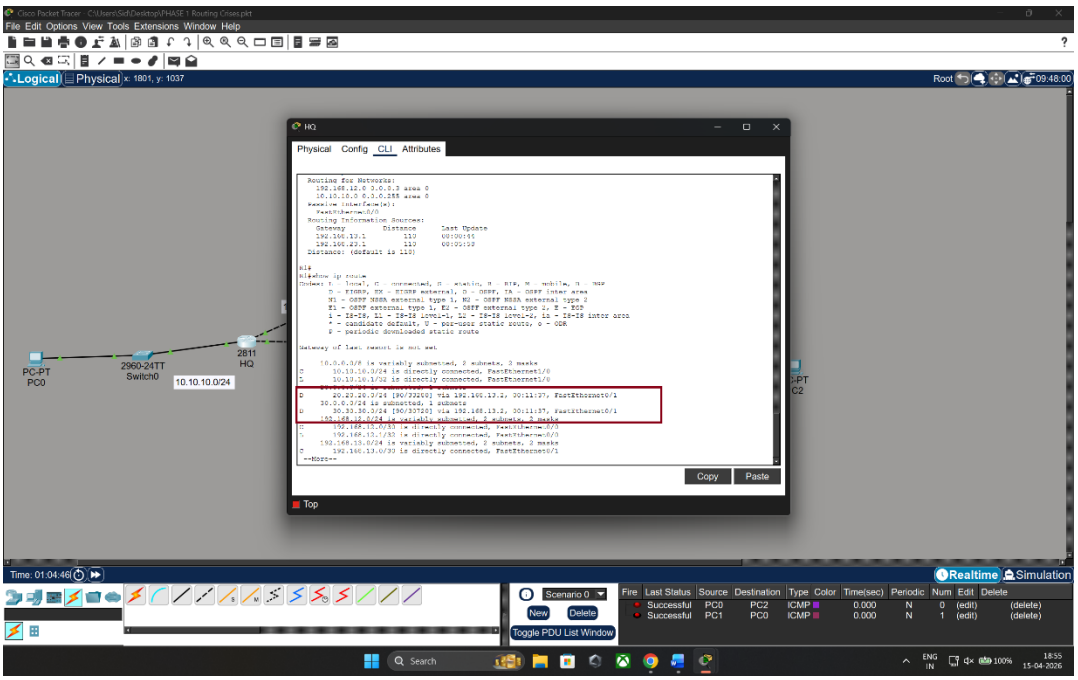
- show ip ospf neighbor : The neighbor table was empty. This confirmed that the adjacency between R1 and R2 had failed, even though the physical link was showing "Green" in Packet Tracer.



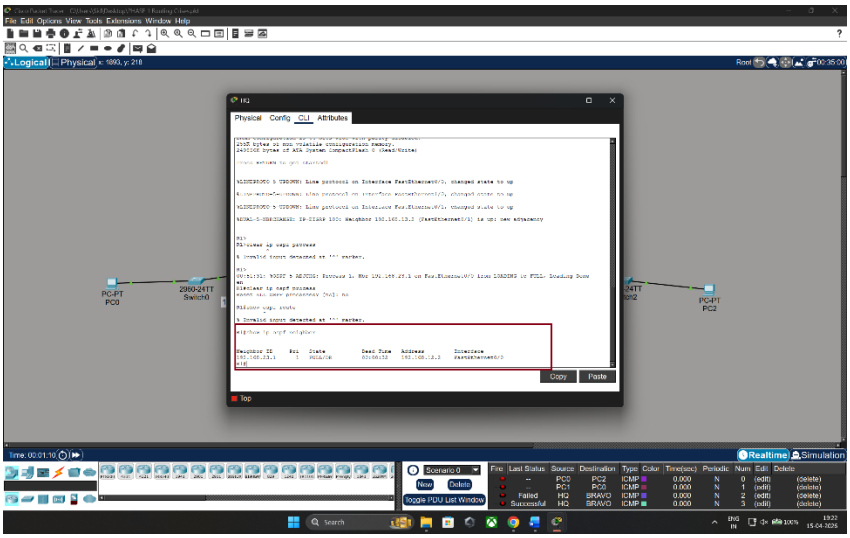
- show ip protocols: Under the OSPF 1 section, FastEthernet 0/0 was explicitly listed as a Passive Interface.



- show ip route: The routing table showed that all "O" (OSPF) routes for the 20.20.20.0 network were missing. R1 was only receiving routes from the EIGRP neighbor



- **To Restore Connectivity**
  - To restore the signal and re-establish the OSPF relationship, I removed the passive-interface suppression with the following commands:
    - router ospf 1
    - no passive-interface f0/0
    - end



- **To verify I ran –**

  - show ip route and confirmed that the "O" routes for the Bravo sector reappeared in the table.End-to-End Connectivity
  - I performed a successful ping from PC Alpha (10.10.10.2) to PC Bravo (20.20.20.2).